

Assignmnet- 10

1. Set up Firebase in a new React app by creating a firebase.js file and initializing Firebase with your project's config details.
2. Create a SignUp component that lets users register with email and password using Firebase Authentication's createUserWithEmailAndPassword method.
3. Build a Login component that authenticates users with email and password using Firebase Authentication's signInWithEmailAndPassword method and displays an error message if login fails.
4. Display the currently logged-in user's email at the top of your app by listening to Firebase Auth state changes with onAuthStateChanged.
5. Use ChatGPT to generate a function that logs out the current user using Firebase Authentication, then implement and test it in your React app.

Hint: Ask ChatGPT for a 'React logout function using Firebase Auth' and adapt the code to your project.

Answer:-

Firebase.js

```
import { initializeApp } from "firebase/app";
import { getAuth } from "firebase/auth";

const firebaseConfig = {
  apiKey: "AIzaSyDuzrblSgnCQ_n0HJUKcYtPNz-1k5rsobo",
  authDomain: "react-firebase-3a941.firebaseio.com",
  projectId: "react-firebase-3a941",
  storageBucket: "react-firebase-3a941.firebaseio.com",
  messagingSenderId: "744439850234",
  appId: "1:744439850234:web:8a54a8b2c01532b111d543",
  measurementId: "G-QYESDFS1GQ"
};

const app = initializeApp(firebaseConfig);

export const auth = getAuth(app);

export default app;
```

SignUp.jsx

```
import { createUserWithEmailAndPassword } from "firebase/auth";
import { useState } from "react";
```

```

import { auth } from "../firebase";

const SignUp = () => {

  const handleSignup = async () => {
    try {
      const userCredential = await createUserWithEmailAndPassword(auth,
email, password);
    } catch (error) {
      alert(error.message)
    }
  }

  const [email, setEmail] = useState("");
  const [password, setPassword] = useState("")
  return (
    <div>
      <h2>Sign Up page</h2>
      <input type="text"
        placeholder="Enter your Email"
        value={email}
        onChange={(e) => { setEmail(e.target.value); }}
      />
      <br /><br />
      <input type="password"
        placeholder="Enter Your Password"
        value={password}
        onChange={(e) => { setPassword(e.target.value) }}
      />
      <br /><br />
      <button onClick={handleSignup}>submit</button>
    </div>
  )
}

export default SignUp;

```

Login.jsx

```

import { signInWithEmailAndPassword } from 'firebase/auth';
import { useState } from 'react';
import { auth } from '../firebase';

const Login = () => {
  const [email, setEmail] = useState("");
  const [password, setPassword] = useState("");

```

```

const [error, setError] = useState("");

const handleLogin = async () => {
  try {
    setError("");
    await signInWithEmailAndPassword(auth, email, password);
    alert("Login succesfully");
  } catch (error) {
    setError("Invalid Email and Password..");
  }
}

return (
  <div>
    <h2>Login page</h2>

    <input type="text"
      placeholder='Enter Your Email'
      value={email}
      onChange={(e) => { setEmail(e.target.value) }}
    />
    <br /><br />
    <input type="text"
      placeholder='Enter Your Password'
      value={password}
      onChange={(e) => { setPassword(e.target.value) }}
    />
    <br /><br />
    <button onClick={handleLogin}>Sign-In</button>
    <br /><br /><br />
    <p style={{ color: "red" }}>{error}</p>
  </div>
)
}

export default Login

```

App.jsx

```

import { onAuthStateChanged, signOut } from 'firebase/auth';
import { useEffect, useState } from 'react';
import { auth } from '../firebase';
import Login from './Login';
import SignUp from './SignUp';

const App = () => {

```

```

const [user, setUser] = useState("");

useEffect(() => {
  const unsubscribe = onAuthStateChanged(auth, (currentUser) => {
    setUser(currentUser);
  })
  return () => unsubscribe();
}, [])

const handleLogout = async() =>{
  try{
    await signOut(auth);
  }catch(error){
    alert(error.message);
  }
}

return (
  <div>
    <h2>FireBase Authentication</h2>
    {
      user ? (<h3>Logged In User : {user.Email}</h3>) : (<h3>No User
Found...</h3>)
    }
    <SignUp />
    <hr />
    <Login />
    {
      user && (
        <button onClick={handleLogout}> Logout</button>
      )
    }
  </div>
)
}

export default App

```